Neural Collapse in Cumulative Link Models for Ordinal Regression: An Analysis with Unconstrained Feature Model

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Abstract

A phenomenon known as "Neural Collapse (NC)" in deep classification tasks, in which the penultimate-layer features and the final classifiers exhibit an extremely simple geometric structure, has recently attracted considerable attention, with the expectation that it can deepen our understanding of how deep neural networks behave. The Unconstrained Feature Model (UFM) has been proposed to explain NC theoretically, and there emerges a growing body of work that extends NC to tasks other than classification and leverages it for practical applications. In this study, we investigate whether a similar phenomenon arises in deep Ordinal Regression (OR) tasks, via combining the cumulative link model for OR and UFM. We show that a phenomenon we call Ordinal Neural Collapse (ONC) indeed emerges and is characterized by the following three properties: (ONC1) all optimal features in the same class collapse to their within-class mean when regularization is applied; (ONC2) these class means align with the classifier, meaning that they collapse onto a one-dimensional subspace; (ONC3) the optimal latent variables (corresponding to logits or preactivations in classification tasks) are aligned according to the class order, and in particular, in the zero-regularization limit, a highly local and simple geometric relationship emerges between the latent variables and the threshold values. We prove these properties analytically within the UFM framework with fixed threshold values and corroborate them empirically across a variety of datasets. We also discuss how these insights can be leveraged in OR, highlighting the use of fixed thresholds.

1 Introduction

In classification tasks on balanced datasets, it has been observed that, after sufficient training, the outputs (or features) of the penultimate layer and the final classifier weights in sufficiently expressive Deep Neural Networks (DNNs) exhibit a remarkably simple symmetric structure. Papyan et al. [2020] conducted thorough experiments across architectures and datasets to corroborate this phenomenon, and identified its four intertwined signatures, which are summarized as **Neural Collapse (NC)**: (NC1)

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all feature vectors from the same class converge exactly onto their shared class mean, extinguishing within-class variance; (NC2) once these class means are recentered at the global mean, they occupy the vertices of a maximally symmetric Simplex Equiangular Tight Frame (Simplex ETF); (NC3) each classifier weight vector becomes parallel to its corresponding class mean vector, reflecting a self-dual alignment; (NC4) the network simply classifies by nearest class mean. NC, being considered to offer a valuable clue toward understanding DNNs, has inspired a number of theoretical studies [Mixon et al., 2022, Fang et al., 2021, Zhu et al., 2021, E and Wojtowytsch, 2020, Lu and Steinerberger, 2022] which introduced **Unconstrained Feature Model (UFM)** being a central theoretical tool in this context.

UFM has allowed extending NC to broader problem settings and facilitated the analytical investigation of its properties [Zhou et al., 2022, Thrampoulidis et al., 2022, Dang et al., 2023, 2024, Li et al., 2024a]. For example, Andriopoulos et al. [2024] applied this framework to multivariate regression, finding a phenomenon called Neural Regression Collapse (NRC) in which features collapse to a target subspace and weight vectors align accordingly. The proliferation of these extensions suggests that NC is a universal phenomenon in DNNs.

Beyond classification and regression, there is a task called **Ordinal Regression (OR)** which aims to solve prediction tasks where labels are discrete categories with a natural order. Unlike classification, not all misclassifications are equally wrong in OR tasks; unlike regression, label values in OR do not bear quantitative information and only their ordering relationship is essential. A common approach to OR problems is to use threshold models [Verwaeren et al., 2012] which assume that an unobserved continuous latent variable generates the observed ordinal response: the map to latent space is traditionally assumed to be linear with respect to (w.r.t.) input datapoints [Herbrich et al., 2000]. Even within this framework, more challenging scenarios can be addressed by applying an appropriate transformation (feature extractor) to the input datapoints, and the effectiveness of DNNs as feature extractors has been reported [Dorado-Moreno et al., 2012, Vargas et al., 2020]. The latent variable in the threshold models occupies the same role as the logit in classification networks, prompting the question of whether phenomena similar to NC may also emerge in the feature space of OR.

To address this question, in this paper we explore phenomena analogous to NC within the context of OR. To that end, we adopt the Cumulative Link Model (CLM) [Agresti, 2010]—a classical type of threshold models—and analyze it in combination with UFM. As a result, we find that such a phenomenon indeed occurs when the ℓ_2 regularization is applied, and we name it Ordinal Neural Collapse (ONC). ONC is characterized by the following three properties:

(ONC1) Within-class Mean Collapse: *all optimal features in the same class collapse to their within-class mean.*

(ONC2) Collapse to One-Dimensional Subspace: *these class means align with the classifier, meaning that they collapse onto a one-dimensional subspace.*

(ONC3) Collapse to Ordinal Structure: *the optimal latent variables are aligned according to the class order, and in particular, in the zero-regularization limit, a highly local and simple relationship emerges between the latent variables and the threshold values.*

We note that this result is obtained under the assumption that all the threshold values of CLM are fixed. Although it is not necessarily a standard assumption in recent studies, we argue that it is necessary for the emergence of ONC and, moreover, can be a practically meaningful assumption. A discussion about this point will be given later in Section 6.

We also validated ONC through experiments using five imbalanced ordinal datasets and a DNN architecture. The result provides clear empirical evidence of ONC under fixed threshold values. Furthermore, our experiments with learnable thresholds still exhibit ONC, implying its robustness.

2 Related work

There already exists a vast body of literature on the extension of NC, and we cite here only recent results that are particularly relevant to the present work.

UFM and related issues. UFM and the related models were proposed almost concurrently in a number of pieces of work [Mixon et al., 2022, Fang et al., 2021, Zhu et al., 2021, E and Wojtowytsch, 2020, Lu and Steinerberger, 2022]. The core idea of UFM lies in decoupling the features from the data by treating the output of a specific layer, typically the penultimate layer, as free learnable variables, while explicitly modeling the nonlinear functions and weight vectors in the subsequent layers. This simplifying assumption enables us to analytically derive nontrivial results also relevant to realistic DNNs. Even in such a simplified model, the analysis can remain nontrivial. For example, UFM generally admits multiple local minima in its loss landscape. Among them, it was shown by Zhu et al. [2021] that only the global minimum exhibits the NC structure in the case of balanced classification. Departing from the typical analysis of the single-layer UFM, the first investigation of UFM with multiple layers was carried out by Tirer and Bruna [2022], finding that NC still emerges as the unique global optimum. Furthermore, to understand the phenomenon called Deep NC, in which NC propagates not only to the final layer but also to intermediate layers [Hui et al., 2022, He and Su, 2023, Rangamani et al., 2023, Parker et al., 2023, Masarczyk et al., 2023], Súkeník et al. [2023] extended UFM to multiple nonlinear layers and proved that, in the case of binary classification, Deep NC emerges as the unique global optimum. Their follow-up study [Súkeník et al., 2024] demonstrated that, in multi-layer architectures, increasing the number of classes causes Deep NC to cease being the optimal solution.

NC extensions under varied settings on classification. In classification, the NC framework has been adapted to different settings from the balanced case. Zhou et al. [2022] showed that cross-entropy, focal loss, label smoothing, and even mean squared error lead to the same NC geometry using UFM. Thrampoulidis et al. [2022] and Hong and Ling [2024] found that with cross-entropy loss and ℓ_2 regularization, class imbalance does not prevent NC1 but the global geometry generalizes from a Simplex ETF to a more general structure. In extremely imbalanced cases, Fang et al. [2021] found a "minority collapse" phenomenon where minority-class features collapse onto a single point. Dang et al. [2023] proved in deep linear UFM that every global minimizer forms orthogonal directions whose magnitudes scale proportionally to class sample sizes, and Dang et al. [2024] proved essentially the same statement for rectified linear unit (ReLU) UFM.

NC extensions beyond classification. Li et al. [2024a] extended NC to multi-label classification by showing that multi-label embeddings lie in the linear span of label-means. Andriopoulos et al. [2024] generalized NC to multivariate regression to find NRC. Wu and Papyan [2024] introduced the concept of "linguistic collapse" in large-scale language models, showing that token embeddings tend toward equal-norm, equal-angle configurations as model scale increases. Furthermore, Nguyen et al. [2024] demonstrated that NC-style embedding collapse also occurs in diffusion models. NC-like phenomena have also been observed in self-supervised learning [Ben-Shaul et al., 2023] and in transfer learning [Galanti et al., 2022, Li et al., 2024b]: the latter studies discussed the relationship between the degree of NC and transfer performance, and also proposed some strategy for leveraging NC insights to improve generalization performance.

3 Formulation

OR. An OR task is formulated as follows. Let $\mathcal{X}$ be an input space and $\mathcal{Y} = \{1, 2, \ldots, Q\}$ be an ordered label set with ordering $1 < 2 < \cdots < Q$. Given a training set $D = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$ with $(\boldsymbol{x}_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, our goal is to learn an order-respecting mapping $r : \mathcal{X} \to \mathcal{Y}$. For each label q, we let $D_q = \{(\boldsymbol{x}_i, y_i) \in D \mid y_i = q\}$ and n_q its size, so that $\sum_{q=1}^{Q} n_q = N$ holds.

CLM. To express the ordinal structure, threshold models introduce a latent variable $z \in \mathbb{R}$ and also a strictly ordered set of "thresholds" $\boldsymbol{b} = (b_0, b_1, \ldots, b_Q)$ which partitions the z-axis. One typically assumes $(b_0, b_Q) = (-\infty, \infty)$ to partition $\mathbb{R}$ properly, and thus each interval is uniquely associated with one category via the decision rule $y = q \iff z \in (b_{q-1}, b_q]$.

In CLMs, the probability of a specific category is expressed through a cumulative probability $P\big(y \leq q \mid z\big)$ conditioned on the latent variable z, which is modeled by using a strictly monotone inverse link function $g : \mathbb{R} \to (0, 1)$ as

$$P\big(y \leq q \mid z\big) = g\big(b_q - z\big). \tag{1}$$

There are several typical choices for g, including the logistic function $g(x) = (1+e^{-x})^{-1}$, the normal cumulative distribution function (CDF) $g(x) = \Phi(x) = \int_{-\infty}^{x} e^{-\frac{1}{2}z^2} dz/\sqrt{2\pi}$, and the Gumbel CDF $g(x) = 1 - e^{-e^{x}}$, which correspond to the logit, probit, and clog-log models, respectively.

An input datapoint $\boldsymbol{x}$ is transformed to a value z in the latent space through a certain map. When using a feature extractor such as DNNs [Vargas et al., 2020], the map is expressed as

$$z = f_{\boldsymbol{w},\theta}(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{h}_\theta(\boldsymbol{x}), \tag{2}$$

where $\boldsymbol{w} \in \mathbb{R}^p$ and $\boldsymbol{h}_\theta : \mathcal{X} \to \mathbb{R}^p$ are the classifier weight vector and feature extractor, respectively. Here, θ represents the parameters of the feature extractor.

Under the model (1), (2), since the probability that y belongs to class q given z is expressed as $P\big(y = q \mid z\big) = P\big(y \le q \mid z\big) - P\big(y \le q-1 \mid z\big)$, the empirical negative log-likelihood given the dataset $\{D_q\}_{q=1}^Q$ becomes

$$\mathcal{L}_{\mathrm{NLL}}(\boldsymbol{w}, \theta, \boldsymbol{b}) = \frac{1}{N} \sum_{q=1}^{Q} \sum_{(\boldsymbol{x}_i, y_i) \in D_q} L(z_i, b_{q-1}, b_q), \qquad z_i = f_{\boldsymbol{w},\theta}(\boldsymbol{x}_i), \tag{3}$$

where we let $L(z, a, b) := -\log[g(b - z) - g(a - z)]$. As in recent practices using DNNs, we consider the ℓ_2 regularization on the parameters:

$$\mathcal{R}(\boldsymbol{w}, \theta) = \tfrac{\lambda_w}{2} \|\boldsymbol{w}\|_2^2 + \tfrac{\lambda_\theta}{2} \|\theta\|_2^2. \tag{4}$$

The overall optimization problem is therefore

$$\min_{\boldsymbol{w},\theta} \big(\mathcal{L}_{\mathrm{NLL}}(\boldsymbol{w}, \theta, \boldsymbol{b}) + \mathcal{R}(\boldsymbol{w}, \theta) \big). \tag{5}$$

UFM for CLM-based OR. In the single-layer UFM, the feature vector $\boldsymbol{h}_\theta(\boldsymbol{x}_i)$ itself is treated as a free learnable variable. As a result, for each datapoint $\boldsymbol{x}_i$, a free variable $\boldsymbol{h}_i$ is associated. For notational simplicity, we relabel this variable as $\boldsymbol{h}_{q,i}$, where q indexes the class and $i = 1, \ldots, n_q$ indexes the datapoints within D_q. The regularization on the parameter θ is assumed to be converted to that on $H := (\boldsymbol{h}_{q,i})_{q,i}$. UFM thus allows us to convert (5) into

$$\min_{\boldsymbol{w},H} \big(\mathcal{L}_{\mathrm{NLL,UFM}}(\boldsymbol{w}, H, \boldsymbol{b}) + \mathcal{R}_{\mathrm{UFM}}(\boldsymbol{w}, H) \big), \tag{6}$$

where

$$\mathcal{L}_{\mathrm{NLL,UFM}}(\boldsymbol{w}, H, \boldsymbol{b}) = \frac{1}{N} \sum_{q=1}^{Q} \sum_{i=1}^{n_q} L(\boldsymbol{w}^\top \boldsymbol{h}_{q,i}, b_{q-1}, b_q), \tag{7}$$

$$\mathcal{R}_{\mathrm{UFM}}(\boldsymbol{w}, H) = \frac{\lambda_w}{2} \|\boldsymbol{w}\|_2^2 + \frac{\lambda_h}{2N} \sum_{q=1}^{Q} \sum_{i=1}^{n_q} \|\boldsymbol{h}_{q,i}\|_2^2. \tag{8}$$

4 Theoretical results based on UFM analysis

Let us state our main theoretical results. Thanks to the structure of our CLM and UFM, (6) can be decomposed into a multi-stage optimization as follows:

$$\min_{w} \left\{ \frac{\lambda_w}{2} w^2 + \frac{1}{N} \sum_{q=1}^{Q} \sum_{i=1}^{n_q} \min_{\boldsymbol{h}_{q,i}} f_q(w, \boldsymbol{h}_{q,i}) \right\}, \tag{9}$$

where

$$f_q(w, \boldsymbol{h}) = L(w \boldsymbol{a}^\top \boldsymbol{h}, b_{q-1}, b_q) + \frac{\lambda_h}{2} \|\boldsymbol{h}\|_2^2, \tag{10}$$

and where $w \ge 0$ and $\boldsymbol{a}$ are the norm of $\boldsymbol{w}$ and the unit vector representing the direction of $\boldsymbol{w}$, respectively, so that $\boldsymbol{w} = w\boldsymbol{a}$ and $\|\boldsymbol{a}\|_2 = 1$ hold. Since our objective function to be minimized in (6) is invariant under any orthogonal transformation $\boldsymbol{w} \to O\boldsymbol{w}, \boldsymbol{h} \to O\boldsymbol{h}, \forall O \in \mathrm{O}(p)$, we can fix the direction $\boldsymbol{a}$ of $\boldsymbol{w}$ without loss of generality. Furthermore, we assume that the derivative g' of the inverse link function g is logarithmically concave (log-concave): some standard choices of g in OR such as the logistic function, the standard normal CDF, and the Gumbel CDF satisfy this assumption.

Under these assumptions, the conditions of ONC can be derived. Before presenting the concrete statements, we first show the following theorem.

Theorem 4.1. *Let $p(x)$ be a log-concave function on $\mathbb{R}$, and let $P(x) = \int_{-\infty}^{x} p(u)\,du$. Then, for any $a < b$, the function $\rho(x) = P(b-x) - P(a-x)$ is log-concave.*

Proof. One can write $\rho(x)$ as

$$\rho(x) = P(b-x) - P(a-x) = \int_{a-x}^{b-x} p(u)\,du = \int_{a}^{b} p(y+x)\,dy, \tag{11}$$

where we let $y = u - x$. Since $p(y+x)$ is log-concave in $\mathbb{R}^2$, one can apply Theorem A.1 with $A = [a,b]$ to conclude that $\rho(x)$ is log-concave in x. □

This means that the log-concavity of g' leads to the convexity of $L(z,a,b) = -\log[g(b-z)-g(a-z)]$ w.r.t. z for any (a,b) satisfying $b > a$. One can further show that, if g' is *strictly* log-concave and positive, then $L(z,a,b)$ is *strictly* convex in z: see Appendix A.2.

We are now ready to state the ONC theorem.

Theorem 4.2 (ONC)**.** *Assume that the inverse link function $g(x)$ defined on $\mathbb{R}$ is differentiable, and that its derivative g' is log-concave. Consider (6) with thresholds $\boldsymbol{b} = (b_0, b_1, \ldots, b_Q)$ satisfying $b_0 < b_1 < \cdots < b_Q$, and let $(\boldsymbol{w}^*, H^*)$ denote the global minimizer. Under the assumption $\lambda_w, \lambda_h > 0$, the following three properties hold:*

(ONC1) *For any class $q \in \mathcal{Y}$, the optimal features $\{\boldsymbol{h}^*_{q,i}\}_i$ in class q become identical:*

$$\boldsymbol{h}^*_{q,i} = \boldsymbol{h}^*_q, \quad \forall i = 1, \ldots, n_q.$$

In other words, the optimal features collapse to their within-class mean $\boldsymbol{h}^*_q$.

(ONC2) *For any class q, the class mean $\boldsymbol{h}^*_q$ becomes parallel to $\boldsymbol{w}^*$, meaning that all class means collapse onto the one-dimensional subspace spanned by $\boldsymbol{w}^*$.*

(ONC3) *The optimal latent variables $z^*_q = (\boldsymbol{w}^*)^\top \boldsymbol{h}^*_q$ satisfy $z^*_1 \leq z^*_2 \leq \cdots \leq z^*_Q$. Moreover, if g' is strictly log-concave and if $\boldsymbol{w}^* \neq \boldsymbol{0}$, then these inequalities hold strictly.*

Proof. By a technical reason, we separately treat the two cases $w^* = 0$ and $w^* \neq 0$, and here provide only the derivation of ONC1 and 2, deferring the proof of ONC3 to Appendix B.
Thanks to the structure of (9), for any fixed w every $\boldsymbol{h}_{q,i}$ can be optimized separately from the other variables, and the objective function is identical for all $i \in \{1, \ldots, n_q\}$. Its explicit form is

$$\arg\min_{\boldsymbol{h}_{q,i}} f_q(w, \boldsymbol{h}_{q,i}) = \arg\min_{\boldsymbol{h}} \left(L(w\boldsymbol{a}^\top \boldsymbol{h}, b_{q-1}, b_q) + \frac{\lambda_h}{2}\|\boldsymbol{h}\|_2^2 \right). \tag{12}$$

First suppose $w \neq 0$. Since $L(z, b_{q-1}, b_q)$ is proven to be convex in z through Theorem 4.1, $L(w\boldsymbol{a}^\top \boldsymbol{h}, b_{q-1}, b_q) =: L_q(w\boldsymbol{a}^\top \boldsymbol{h})$ is also convex in $\boldsymbol{h}$. Since the term $(\lambda_h/2)\|\boldsymbol{h}\|_2^2$ is strictly convex, the total objective function to be minimized is strictly convex w.r.t. $\boldsymbol{h}$. On the other hand, let $\boldsymbol{v}_q$ be the gradient of $L_q(w\boldsymbol{a}^\top \boldsymbol{h})$ at $\boldsymbol{h} = \boldsymbol{0}$. Thanks to the convexity of $L_q(w\boldsymbol{a}^\top \boldsymbol{h})$, one has

$$L_q(w\boldsymbol{a}^\top \boldsymbol{h}) - L_q(0) \geq \boldsymbol{v}_q^\top \boldsymbol{h}, \quad \forall \boldsymbol{h}, \tag{13}$$

which implies that the objective function is bounded from below:

$$L_q(w\boldsymbol{a}^\top \boldsymbol{h}) + \frac{\lambda_h}{2}\|\boldsymbol{h}\|_2^2 \geq L_q(0) + \frac{1}{2}\lambda_h \left\| \boldsymbol{h} + \frac{\boldsymbol{v}_q}{\lambda_h} \right\|_2^2 - \frac{1}{2\lambda_h}\|\boldsymbol{v}_q\|_2^2 \geq L_q(0) - \frac{1}{2\lambda_h}\|\boldsymbol{v}_q\|_2^2 > -\infty. \tag{14}$$

Hence, the strict convexity and the boundedness of $L_q(w\boldsymbol{a}^\top \boldsymbol{h}) + \frac{\lambda_h}{2}\|\boldsymbol{h}\|_2^2$ guarantee the uniqueness of the minimizer, proving ONC1. The proof of ONC2 is more straightforward. Let $\boldsymbol{h}_\parallel$ denote the projection of $\boldsymbol{h}$ on $\boldsymbol{a}$ and $\boldsymbol{h}_\perp = \boldsymbol{h} - \boldsymbol{h}_\parallel$. Then we have

$$L_q(w\boldsymbol{a}^\top \boldsymbol{h}) + \frac{\lambda_h}{2}\|\boldsymbol{h}\|_2^2 = L_q(w\boldsymbol{a}^\top \boldsymbol{h}_\parallel) + \frac{\lambda_h}{2}\|\boldsymbol{h}_\parallel\|_2^2 + \frac{\lambda_h}{2}\|\boldsymbol{h}_\perp\|_2^2. \tag{15}$$

Hence, the minimization of this w.r.t. $\boldsymbol{h}_\perp$ yields $\boldsymbol{h}^*_\perp = \boldsymbol{0}$, showing ONC2.
Next we assume $w = 0$. In this case, the dependence of the objective function on $\boldsymbol{h}_{q,i}$ only appears in the regularization term and the optimization thus yields $\boldsymbol{h}^*_{q,i} = \boldsymbol{0}$ for all q, i. Hence, the ONC properties appear trivially. □

In contrast to ONC1 and 2, which only require the convexity of $L(z, a, b)$, ONC3 has a more quantitative information about the problem. Actually, the values of $w^*, \boldsymbol{z}^*$ are determined from a set of equations deduced from the stationarity condition of (9). Borrowing the terminology from statistical physics, we call this set of equations **Equations Of State (EOS)**. Analyzing EOS leads to a derivation of ONC3, but it is involved and is deferred to Appendix B. The solution of EOS exhibits some singularity at certain parameter values, and also some simple behaviors in certain limits. The next theorem summarizes these findings.

Theorem 4.3 (EOS, phase transition, and some limiting behaviors)**.** *Consider the same situation as in Theorem 4.2. If the optimal norm value satisfies $w^* > 0$, w^* and the optimal latent variables $\boldsymbol{z}^*$ obey the following set of equations which we call EOS:*

$$\frac{g'(b_q - z_q^*) - g'(b_{q-1} - z_q^*)}{g(b_q - z_q^*) - g(b_{q-1} - z_q^*)} + \lambda_h \frac{z_q^*}{(w^*)^2} = 0, \quad q = 1, \ldots, Q, \tag{16a}$$

$$\lambda_w w^* - \frac{\lambda_h}{(w^*)^3} \sum_{q=1}^{Q} \alpha_q (z_q^*)^2 = 0, \tag{16b}$$

where $\alpha_q = n_q/N$. Additionally assuming the continuity and monotonicity of w^* w.r.t. λ_h and λ_w, this EOS implies a phase transition with the phase boundary in the (λ_h, λ_w)-plane characterized by

$$\lambda_h \lambda_w = C := \sum_{q=1}^{Q} \alpha_q \left(\frac{g'(b_q) - g'(b_{q-1})}{g(b_q) - g(b_{q-1})} \right)^2. \tag{17}$$

Namely, for $\lambda_h \lambda_w \geq C$ the trivial solution $w^* = 0, \boldsymbol{z}^* = \boldsymbol{0}$ becomes the optimal solution to (9), while for $\lambda_h \lambda_w < C$ the nontrivial solution $w^* > 0, \boldsymbol{z}^* \neq 0$, which obey EOS, becomes the optimal one.

Moreover, this EOS admits a simple behavior emerging in the limit where the product $\lambda_h \lambda_w$ approaches zero. In that limit, $\boldsymbol{z}^*$ is determined by

$$g'(b_q - z_q^*) = g'(b_{q-1} - z_q^*), \quad q = 1, \ldots, Q, \tag{18}$$

and one has $w^* = \Theta\big((\lambda_h/\lambda_w)^{1/4}\big)$.

Proof. Applying ONC2, we have $z_q = w\boldsymbol{a}^\top \boldsymbol{h}_q$. Thus we may rewrite the squared norm $\|\boldsymbol{h}_q\|_2^2$ as z_q^2/w^2 and the optimization w.r.t. $\boldsymbol{w}$ and $(\boldsymbol{h}_q)_q$ in (9) can be reduced to those w.r.t. $w, \boldsymbol{z}$. Taking the stationarity condition w.r.t. $\boldsymbol{z}$ and w lead to EOS.

Next, we examine the phase transition and the phase boundary. One subtlety in analyzing the nature of the phase transition is that the trivial solution $(w^*, \boldsymbol{z}^*) = (0, \boldsymbol{0})$ does not satisfy EOS within the whole parameter region where it is optimal. However, thanks to the assumed continuity and monotonicity of w^*, exactly on the phase boundary the trivial solution must satisfy EOS. Therefore, we search for a condition under which EOS admits the trivial solution. By substituting (16a) into (16b) to eliminate the explicit dependence on w^*, we obtain the following equation:

$$\lambda_w = \frac{1}{\lambda_h} \sum_{q=1}^{Q} \alpha_q \left(\frac{g'(b_q - z_q^*) - g'(b_{q-1} - z_q^*)}{g(b_q - z_q^*) - g(b_{q-1} - z_q^*)} \right)^2. \tag{19}$$

Inserting the trivial solution $\boldsymbol{z}^* = \boldsymbol{0}$ into this leads to the phase boundary (17). Thanks to the assumed monotonicity, once the solution becomes the trivial one, it continues to be so above the boundary.

Finally, the limiting behavior is investigated. By inserting (16b) into (16a) to eliminate the explicit dependence on w^*, we have

$$\frac{g'(b_q - z_q^*) - g'(b_{q-1} - z_q^*)}{g(b_q - z_q^*) - g(b_{q-1} - z_q^*)} + \sqrt{\lambda_w \lambda_h} \frac{z_q^*}{\sqrt{\sum_{q'=1}^{Q} \alpha_{q'} (z_{q'}^*)^2}} = 0, \quad q = 1, \ldots, Q. \tag{20}$$

This yields (18) when $\lambda_h \lambda_w$ is sent to zero, as long as $\boldsymbol{z}^* \neq \boldsymbol{0}$. Then, solving (16b) w.r.t. w^*, we have the scaling $w^* = \Theta\big((\lambda_h/\lambda_w)^{1/4}\big)$. □

The scaling $w^* = \Theta\big((\lambda_h/\lambda_w)^{1/4}\big)$ means that w^* in the limit λ_h and/or $\lambda_w \to 0$ may diverge, vanish, or remain finite depending on how one takes the limit.

Equation (18) in the vanishing regularization limit is fairly striking since it provides a simple local relation between $\boldsymbol{b}$ and $\boldsymbol{z}^*$. Especially, if the inverse link function satisfies a symmetry $1 - g(x) = g(-x)$, which is the case for the logit and probit models, one has $g'(x) = g'(-x)$, and (18) thus implies

$$z_q^* = \frac{b_q + b_{q-1}}{2}. \tag{21}$$

This simple relation will be verified later in experiments using real-world datasets.

For illustration, we numerically solved (16) for the logit model and plotted the solution in Fig. 1. The

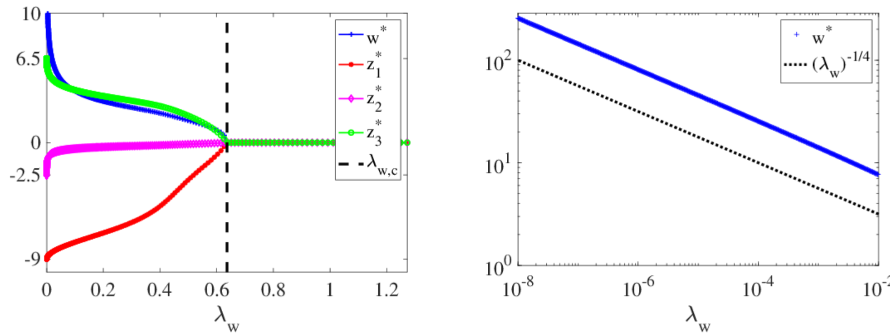

Figure 1: Solution behavior of EOS in the logit model for $Q = 3$ with $\boldsymbol{b} = (-10, -8, 3, 10)$ at $\lambda_h = 1$. (Left) w^* and $\boldsymbol{z}^*$ are plotted against λ_w on a linear scale. A clear phase transition appears at $\lambda_{w,c} = C/\lambda_h$ (vertical broken line), and the values of $\boldsymbol{z}^*$ in the limit $\lambda_w \to 0$ match well with the theoretical prediction ($z_q^* = (b_q + b_{q-1})/2$). (Right) w^* is plotted on a log-log scale in the small-λ_w region. A power-law divergence with exponent $-1/4$, corresponding to the scaling $w^* = \Theta\big((\lambda_h/\lambda_w)^{1/4}\big)$ with fixed λ_h, is clearly observed.

analytical prediction about the critical point and the limiting behaviors were certainly reproduced.

5 Experiments

5.1 Experimental setting

Inverse link functions. Two symmetric inverse link functions, the logistic function and the normal CDF, which correspond to the logit and probit models, respectively, were treated in the experiment.

Datasets and neural networks. We used five OR datasets with the largest number of data points—ER, LE, SW, CA, and WR—among those publicly available from Gutiérrez et al. [2016]. Thirty pre-defined training–validation splits with identical label distributions have been officially released, and we used them as-is in this study. As the neural network architecture, we employed a multilayer perceptron with residual connections. The weight decay was set to a small value 5×10^{-3}. The motivation for this setting is that, in the small-regularization limit, a very simple result emerges as shown in (18), which facilitates experimental verification. Further details are provided in Appendix C.

Treatment of thresholds. We considered two cases, fixed and learnable thresholds, in the experiment.

For the fixed case, to ensure that the ignored tail probabilities are sufficiently small, the edge thresholds b_0 and b_Q were symmetrically fixed ($b_0 = -b_Q$) to sufficiently large values and the remaining thresholds were evenly spaced over the interval $[b_0, b_Q]$. Under this setting, we solved (5), where θ denotes the DNN parameters.

In the learnable case, we set $b_0 = -\infty$ and $b_Q = +\infty$ and learned b_q with $q = 1, \ldots, Q-1$. To guarantee the strict ordering between the threshold values, we parameterized them with $\boldsymbol{s} \in \mathbb{R}^{Q-1}$ as

$$b_q(\boldsymbol{s}) := \sum_{j=1}^{q} \log\left(1 + e^{s_j}\right) - \frac{1}{Q-1} \sum_{j=1}^{Q-1} \log\left(1 + e^{s_j}\right), \quad q = 1, \ldots, Q-1. \tag{22}$$

Correspondingly, we solved $\min_{\boldsymbol{w},\theta,\boldsymbol{s}} \left(\mathcal{L}_{\mathrm{NLL}}(\boldsymbol{w}, \theta, \boldsymbol{b}(\boldsymbol{s})) + \mathcal{R}(\boldsymbol{w}, \theta)\right)$ instead of (5).

Evaluation metrics. We used two basic training metrics for evaluation: $\mathcal{L}_{\mathrm{NLL}}$ and the mean absolute error (MAE) for label prediction, $\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left|\hat{y}_i - y_i\right|$ where $\hat{y}_i$ is the predicted label. Besides, let $\boldsymbol{h}_\theta(\cdot)$ denote the penultimate-layer output of our DNN, and let $\bar{\boldsymbol{h}}_q = (1/n_q) \sum_{(\boldsymbol{x}_i, y_i) \in D_q} \boldsymbol{h}_\theta(\boldsymbol{x}_i)$ and $\bar{\boldsymbol{h}} = (1/N) \sum_{i=1}^{N} \boldsymbol{h}_\theta(\boldsymbol{x}_i)$ represent the class-wise and global feature means, respectively. Using these, we introduced the following four quantitative indicators for ONC:

$$\underline{\mathrm{ONC}_1} = \frac{(1/Q) \sum_{q=1}^{Q} \frac{1}{N_q} \sum_{(\boldsymbol{x}_i, y_i) \in D_q} \|\boldsymbol{h}_\theta(\boldsymbol{x}_i) - \bar{\boldsymbol{h}}_q\|_2}{(1/N) \sum_{i=1}^{N} \|\boldsymbol{h}_\theta(\boldsymbol{x}_i) - \bar{\boldsymbol{h}}\|_2}, \tag{23}$$

$$\underline{\mathrm{ONC}_{2\text{-}1}} = \frac{\sum_{q=1}^{Q} \|(\bar{\boldsymbol{h}}_q - \bar{\boldsymbol{h}}) - \left(\boldsymbol{u}^\top (\bar{\boldsymbol{h}}_q - \bar{\boldsymbol{h}})\right) \boldsymbol{u}\|_2^2}{\sum_{q=1}^{Q} \|\bar{\boldsymbol{h}}_q - \bar{\boldsymbol{h}}\|_2^2}, \quad \underline{\mathrm{ONC}_{2\text{-}2}} = 1 - \left|\frac{\boldsymbol{w}^\top \boldsymbol{u}}{\|\boldsymbol{w}\|_2}\right|, \tag{24}$$

$$\underline{\mathrm{ONC}_3} = \frac{\sum_{q=1}^{Q-1} |b_q - (z_q + z_{q+1})/2|}{\sum_{q=1}^{Q-1} (b_{q+1} - b_q)}, \tag{25}$$

where $\boldsymbol{u}$ is the unit first principal component of $\{\bar{\boldsymbol{h}}_q - \bar{\boldsymbol{h}}\}_{q=1}^{Q}$. $\underline{\mathrm{ONC}_1}$ is the indicator for ONC1 and becomes zero when ONC1 exactly happens; $\underline{\mathrm{ONC}_{2-1}}$ quantifies whether each class mean collapses onto the dominant one-dimensional subspace represented by $\boldsymbol{u}$, while $\underline{\mathrm{ONC}_{2-2}}$ measures whether $\boldsymbol{w}$ also collapses onto the same subspace; since our experiments focus on the small-regularization regime under the use of symmetric g, ONC3 is expected to emerge in the form (21), and accordingly $\underline{\mathrm{ONC}_3}$ serves as an appropriate indicator for it.

5.2 Results

In the main text, we present only the results for the ER dataset using the logit model. Experiments conducted under different settings also yielded consistent results, which are reported in Appendix D.

Figure 2 plots the evolution of all the evaluation metrics. Both training and validation MAE

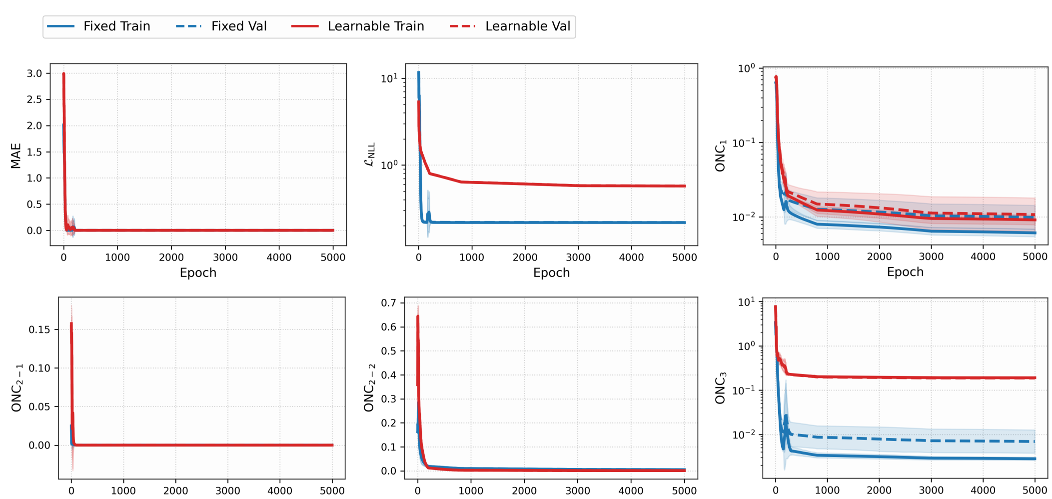

Figure 2: Epoch-wise average metrics curves for the ER dataset with the logit model.

approached zero, indicating that all samples were correctly classified. $\underline{\mathrm{ONC}_{2\text{-}1}}$ and $\underline{\mathrm{ONC}_{2\text{-}2}}$ also rapidly approached zero, showing that the feature vectors collapsed onto the one-dimensional subspace spanned by $\boldsymbol{w}$. As training proceeded, $\overline{\mathrm{ONC}_1}$ decreased steadily, confirming that features collapsed toward their class means. For $\overline{\mathrm{ONC}_3}$ we observed a clear difference between the two threshold strategies. With fixed thresholds, $\overline{\mathrm{ONC}_3}$ took a small value from an early stage and continued to decrease as training progressed. In contrast, with learnable thresholds, $\overline{\mathrm{ONC}_3}$ seemed to converge to a non-zero value. These observations indicate that the simple form of ONC3, given by (21), practically holds under fixed thresholds but does not hold under learnable ones.

Figure 3 illustrates the evolution of the latent and the PCA-projected feature spaces. The red dashed

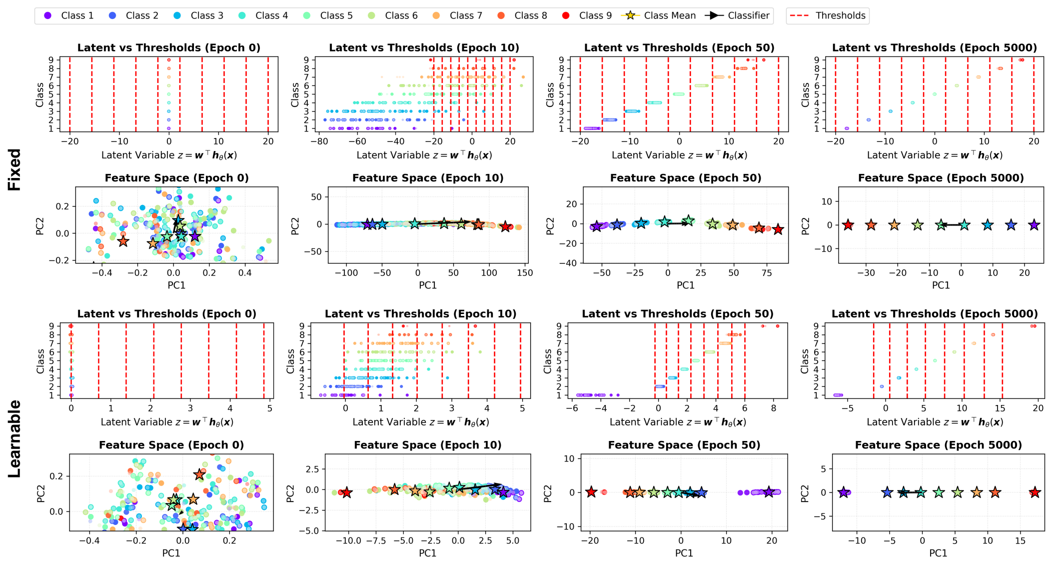

Figure 3: Visualization of the latent and feature spaces for the ER dataset using the logit model.

lines denote the thresholds. Feature points are color-coded by class, with validation features in lighter shades. Class means are highlighted with star markers, and the black arrow indicates the classifier weight. This visualization clearly reveal the emergence of ONC1–3.

6 Discussion

Perspectives and future directions. ONC can also offer practical advantages. Although fixed thresholds were introduced primarily to establish ONC, as shown in Fig. 2, they tended to yield faster and more stable convergence compared with learnable thresholds. It is worthwhile to investigate whether this advantage generalizes to more challenging datasets or more complex DNN architectures.

Moreover, in tasks with label imbalance, it is well known that the performance for minority classes often degrades. Since fixed thresholds provide a fairer allocation of the latent space—and hence of predicted probabilities—across all classes, they may offer greater robustness and generalization under label imbalance or label shift, compared with the learnable thresholds. Although this hypothesis can be tested experimentally, the datasets used in this study were relatively simple and yielded near-zero validation MAE. As a result, we were unable to evaluate this effect in the current setting. We leave this as an important direction for future work.

Furthermore, the geometric structure induced by ONC can be directly utilized in the design of regularization terms or loss functions. For instance, adding lightweight penalties that attract each class mean toward the classifier axis or to the corresponding threshold midpoint may accelerate training, especially in scenarios with scarce labels or significant class imbalance. We also leave such extensions as promising directions for future exploration.

Limitations. The theoretical development in Section 4 assumes that the thresholds $\boldsymbol{b}$ are fixed. Although our experimental results suggest that ONC also emerges when $\boldsymbol{b}$ is learnable, this has not yet been theoretically established. Moreover, we believe that there exist only two phases—one with the trivial solution and the other with a non-trivial solution—but we have not succeeded in rigorously proving this. Instead, in Theorem 4.3, we circumvented this gap by assuming the continuity and monotonicity of w^* w.r.t. λ_w and λ_h.

Conclusion. This study extended the NC theory to CLM-based OR. We proved under the UFM framework that three hallmark properties of ONC—within-class mean collapse, alignment of class means along a one-dimensional classifier subspace, and naturally ordered placement of latent variables—inevitably emerge. Furthermore, in the small-regularization limit, we showed that a highly local and simple relation between the thresholds and the latent variables emerges. Experiments on real-world datasets confirmed the theoretical predictions and additionally demonstrated that fixed thresholds not only preserve the ONC structure but also accelerate and stabilize training. These make ONC a potentially effective concept for improving practical performance in generic OR tasks.

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A Convexity

A.1 Prékopa's theorem

We first recall a theorem, due to Prékopa [1973], which is derived from the Prékopa-Leindler inequality on log-concave distributions. It basically states that marginalization preserves the log-concavity. We use it to prove Theorem 4.1 in the main text.

Theorem A.1 (Theorem 6 of Prékopa [1973])**.** *Let $f(\boldsymbol{x}, \boldsymbol{y})$ be a function of $(n+m)$ variables where $\boldsymbol{x}$ and $\boldsymbol{y}$ are n- and m-dimensional, respectively. Suppose that f is log-concave on $\mathbb{R}^{n+m}$ and let A be a convex subset of $\mathbb{R}^m$. Then the function of $\boldsymbol{x}$ defined by*

$$\int_A f(\boldsymbol{x}, \boldsymbol{y})\, d\boldsymbol{y} \tag{26}$$

is log-concave on $\mathbb{R}^n$.

A.2 Strict log-concavity

Here we show that the strict log-concavity of $g'(x)$, as well as the condition $g'(x) > 0$ for all $x \in \mathbb{R}$, ensures the strict convexity of $L(z, a, b)$ in z. Although proving it would be easy if one can assume differentiability of g', as demonstrated in Appendix B, it holds even without the differentiability assumption, as shown in the following.

The following theorem, as well as the proof, was inspired by the argument in Barbato's article [Barbato (https://math.stackexchange.com/users/86335/maurizio-barbato), 2018]. The first half of the proof basically follows that of [Brascamp and Lieb, 1975, Theorem 1.1]. On the other hand, the latter half of the proof is different from Barbato's.

Theorem A.2. *Let $p(x)$ be a positive, bounded, and* strictly *log-concave function on $\mathbb{R}$, and let $P(x) = \int_{-\infty}^{x} p(u)\, du$. Then the function $\rho(x, y) = P(x) - P(y)$ defined on $\{(x, y) \in \mathbb{R}^2 \mid x > y\}$ is* strictly *log-concave.*

Proof. Choose arbitrary (x, y) and (x', y') satisfying $x > y$, $x' > y'$, and $(x, y) \neq (x', y')$, and let

$$Q(\lambda, u) = \mathbb{1}(\lambda x + (1-\lambda)x' > u > \lambda y + (1-\lambda)y')p(u), \quad (\lambda, u) \in [0,1] \times \mathbb{R}, \tag{27}$$

where $\mathbb{1}(\cdot)$ is the indicator function. The function $Q(\lambda, u)$ is log-concave in (λ, u) because it is a product of log-concave functions. For any $\lambda \in [0, 1]$, one has $0 < \sup_{u \in \mathbb{R}} Q(\lambda, u) < \infty$ because $Q(\lambda, u) \leq p(u) < \infty$ and $p(u) > 0$. Let $\bar{Q}(\lambda, u) := e^{b\lambda} Q(\lambda, u)$ with $b := \log \frac{\sup_{u \in \mathbb{R}} Q(0,u)}{\sup_{u \in \mathbb{R}} Q(1,u)}$, so that $\bar{z} := \sup_{u \in \mathbb{R}} \bar{Q}(1, u) = \sup_{u \in \mathbb{R}} \bar{Q}(0, u) < \infty$. The function $\bar{Q}(\lambda, u)$ is also log-concave in (λ, u). We let

$$\begin{aligned} R(\lambda) &:= \int_{-\infty}^{\infty} Q(\lambda, u)\, du \\ &= P(\lambda x + (1-\lambda)x') - P(\lambda y + (1-\lambda)y') \\ &= \rho(\lambda x + (1-\lambda)x', \lambda y + (1-\lambda)y') \end{aligned} \tag{28}$$

and

$$\bar{R}(\lambda) := \int_{-\infty}^{\infty} \bar{Q}(\lambda, u)\, du = e^{b\lambda} R(\lambda). \tag{29}$$

We will prove the strict inequality

$$\bar{R}(\lambda) > \lambda \bar{R}(0) + (1-\lambda)\bar{R}(1), \quad \forall \lambda \in (0, 1), \tag{30}$$

because it implies the strict concavity of $\rho(x, y)$. Since every positive, strict concave function is strictly log-concave, it proves the theorem.

For $z > 0$, let

$$C(z) = \{(\lambda, u) \in [0,1] \times \mathbb{R} \mid \bar{Q}(\lambda, u) \geq z\} \tag{31}$$

be the z-superlevel set of $\bar{Q}(\lambda, u)$, and

$$C(\lambda, z) = \{u \in \mathbb{R} \mid \bar{Q}(\lambda, u) \geq z\}. \tag{32}$$

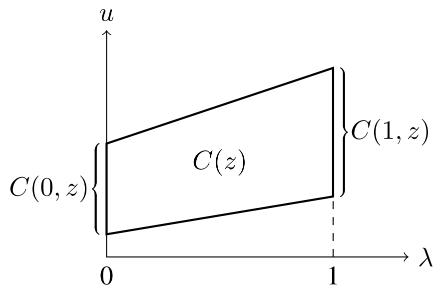

Figure 4: The set $C(z)$ under the equality. The set $C(z)$ should be a trapezoid with $C(0,z)$ and $C(1,z)$ its parallel sides.

be the z-superlevel set of $\bar{Q}(\lambda,u)$ for λ fixed. These sets are convex because of the log-concavity of $\bar{Q}$. In particular, if $z\in(0,\bar{z}]$, $C(z)$ is nonempty, and $C(\lambda,z)$ is a line segment. Let $g(\lambda,z)$ be the length of the line segment $C(\lambda,z)$, and $g(\lambda,z)=0$ if $C(\lambda,z)$ is empty. Then for any $z\in(0,\bar{z})$ the function $g(\lambda,z)$ is positive and concave in λ due to the convexity of $C(z)$, implying that the inequality

$$g(\lambda,z)\geq\lambda g(0,z)+(1-\lambda)g(1,z) \tag{33}$$

holds. One has

$$\begin{aligned}\bar{R}(\lambda)&=\int_0^\infty\mu(\{u\mid\bar{Q}(\lambda,u)>z\})\,dz\\&=\int_0^\infty\mu(\{u\mid\bar{Q}(\lambda,u)\geq z\})\,dz\\&=\int_0^\infty g(\lambda,z)\,dz,\end{aligned} \tag{34}$$

where μ is the Lebesgue measure on $\mathbb{R}$, and therefore the inequality

$$\bar{R}(\lambda)\geq\lambda\bar{R}(0)+(1-\lambda)\bar{R}(1), \tag{35}$$

holds for any $\lambda\in[0,1]$.

In the following, we will prove the strict inequality (30). For this purpose, assume, to the contrary, that there exists $\lambda\in(0,1)$ such that the equality $\bar{R}(\lambda)=\lambda\bar{R}(0)+(1-\lambda)\bar{R}(1)$ holds. It implies that for that value of λ we have equality in (33) almost everywhere in z, which means that it holds for every z because of the continuity of $g(\lambda,z)$ in z. Furthermore, due to the concavity of $g(\lambda,z)$ in λ, the equality in (33) should hold for all $\lambda\in[0,1]$. The argument so far has shown that for all $z\in(0,\bar{z}]$ the set $C(z)$ is the trapezoid with $C(0,z)$ and $C(1,z)$ its parallel sides (Fig. 4). The two remaining sides of the trapezoid are not perpendicular to the λ-axis. Note also that $C(0,z)\subset[x,y]$ and $C(1,z)\subset[x',y']$ hold.

We first assume $b\neq0$. Take any u_0 in the interval (x,y), and let $z_0=p(u_0)\in(0,\bar{z}]$. One can then take a neighborhood of $(\lambda,u)=(0,u_0)$ in $[0,1]\times\mathbb{R}$ where $\bar{Q}(\lambda,u)=e^{b\lambda}p(u)$ holds. The boundary of the z_0-superlevel set $C(z_0)$ not perpendicular to the λ-axis in such a neighborhood is given by

$$\log p(u)+b\lambda=\log z_0. \tag{36}$$

As we have assumed $b\neq0$, this coincides with a straight line in the (λ,u)-plane only if $\log p(u)$ is linear in u, which contradicts the assumption that $p(u)$ is strictly log-concave.

We next assume $b=0$. As above, take any u_0 in the interval (x,y) and let $z_0=p(u_0)$. The boundary of the z_0-superlevel set $C(z_0)$ not perpendicular to the λ-axis in a neighborhood of $(\lambda,u)=(0,u_0)$ is given by $\log p(u)=\log z_0$, which is equivalent to $u=u_0$. The line segment $\{(\lambda,u)\in[0,1]\times\{u_0\}\}$ should form a side of the trapezoid $C(z_0)$, which requires $u_0\in[x',y']$ to hold. We have thus shown the inequalities $x\geq x'$ and $y\leq y'$. The above argument can be repeated by swapping the role of (x,y) and (x',y'), showing $x\leq x'$ and $y\geq y'$. One could then conclude that $(x,y)=(x',y')$, which contradicts the assumption $(x,y)\neq(x',y')$.

We have so far shown that assuming the equality $\bar{R}(\lambda) = \lambda\bar{R}(0) + (1-\lambda)\bar{R}(1)$ leads to contradiction, thus proving that $\bar{R}(\lambda)$ satisfies the strict inequality (30). One thus has

$$\bar{R}(\lambda) > \lambda\bar{R}(0) + (1-\lambda)\bar{R}(1) \geq [\bar{R}(0)]^{\lambda}[\bar{R}(1)]^{1-\lambda}, \tag{37}$$

where the last inequality is due to the inequality of the weighted arithmetic and geometric means. This proves the strict log-concavity of $\rho(x, y)$, thereby proving the theorem. □

It should be noted that the function $Q(\lambda, u)$ defined in (27) is itself log-concave but not strictly log-concave, so that one cannot apply the argument in Prékopa [1973] on the strict log-concavity to our case.

B Derivation of ONC3

We show in this section the ordering $z_1^* \leq z_2^* \leq \cdots \leq z_Q^*$ of the optimal latent variables. For this purpose, we discuss how the minimizer of the optimization problem

$$\min_x \left(L(x, a, b) + \frac{\lambda}{2} x^2 \right), \quad \lambda > 0,\ a < b, \tag{38}$$

with

$$L(x, a, b) = -\log[P(b - x) - P(a - x)] \tag{39}$$

behaves as one changes a, b, where P is an indefinite integral of a function p, which is log-concave on $\mathbb{R}$. It is because the optimal latent variable $z_q^* = w\boldsymbol{a}^\top \boldsymbol{h}_q^*$ given $w > 0$ is determined as the optimal solution of the following minimization:

$$\min_x \left(L(x, b_{q-1}, b_q) + \frac{\lambda_h}{2w^2} x^2 \right). \tag{40}$$

Theorem 4.1 ensures that $L(x, a, b)$ is convex in x for any a, b, which in turn ensures that the minimizer $\hat{x}$ of (38) is unique. We show in the following that the minimizer $\hat{x} = \hat{x}(a, b)$ is monotonically non-decreasing in a and b. This monotonicity will prove the desired ordering of $\{z_q^*\}_q$.

The minimizer $\hat{x}$ satisfies the stationarity condition

$$L_x(\hat{x}, a, b) + \lambda\hat{x} = 0, \tag{41}$$

where the subscript x of L denotes the partial derivative of L with respect to x. It should be noted that particularizing (41) in the optimization of the latent variable z_q yields EOS (16a).

We first assume differentiability of p. Taking the derivative of both sides of (41) w.r.t. a, one has

$$L_{xx}(\hat{x}, a, b)\hat{x}_a + L_{xa}(\hat{x}, a, b) + \lambda\hat{x}_a = 0, \tag{42}$$

yielding

$$\hat{x}_a = -\frac{L_{xa}(\hat{x}, a, b)}{L_{xx}(\hat{x}, a, b) + \lambda}. \tag{43}$$

Similarly, one has

$$\hat{x}_b = -\frac{L_{xb}(\hat{x}, a, b)}{L_{xx}(\hat{x}, a, b) + \lambda}. \tag{44}$$

Since $L(x,a,b)$ is convex in x, one has $L_{xx}(x,a,b) \geq 0$. One also has

$$\begin{aligned}
L_x(x,a,b) &= \frac{p(b-x)-p(a-x)}{P(b-x)-P(a-x)}, \\
L_{xa}(x,a,b) &= -\frac{p'(a-x)}{P(b-x)-P(a-x)} + \frac{[p(b-x)-p(a-x)]p(a-x)}{[P(b-x)-P(a-x)]^2} \\
&= -\frac{p(a-x)}{P(b-x)-P(a-x)}\left[\frac{p'(a-x)}{p(a-x)} - \frac{p(b-x)-p(a-x)}{P(b-x)-P(a-x)}\right], \\
L_{xb}(x,a,b) &= \frac{p'(b-x)}{P(b-x)-P(a-x)} - \frac{[p(b-x)-p(a-x)]p(b-x)}{[P(b-x)-P(a-x)]^2} \\
&= \frac{p(b-x)}{P(b-x)-P(a-x)}\left[\frac{p'(b-x)}{p(b-x)} - \frac{p(b-x)-p(a-x)}{P(b-x)-P(a-x)}\right], \\
L_{xx}(x,a,b) &= -\frac{p'(b-x)-p'(a-x)}{P(b-x)-P(a-x)} + \frac{[p(b-x)-p(a-x)]^2}{[P(b-x)-P(a-x)]^2} \\
&= -[L_{xa}(x,a,b)+L_{xb}(x,a,b)]. \qquad (45)
\end{aligned}$$

Since $p(u)$ is assumed log-concave, $(\log p(u))' = p'(u)/p(u)$ is monotonically non-increasing. One therefore has, via the technique used in [Dierker, 1991, Proof of Lemma 1] and Bagnoli and Bergstrom [2005],

$$\begin{aligned}
\frac{p'(a-x)}{p(a-x)}[P(b-x)-P(a-x)] &= \frac{p'(a-x)}{p(a-x)}\int_{a-x}^{b-x} p(u)\,du \\
&\geq \int_{a-x}^{b-x} \frac{p'(u)}{p(u)}p(u)\,du \\
&= \int_{a-x}^{b-x} p'(u)\,du \\
&= p(b-x)-p(a-x), \qquad (46)
\end{aligned}$$

implying that $L_{xa}(x,a,b) \leq 0$ holds, which in turn proves $\hat{x}_a \geq 0$. Similarly, one has

$$\begin{aligned}
\frac{p'(b-x)}{p(b-x)}[P(b-x)-P(a-x)] &= \frac{p'(b-x)}{p(b-x)}\int_{a-x}^{b-x} p(u)\,du \\
&\leq \int_{a-x}^{b-x} \frac{p'(u)}{p(u)}p(u)\,du \\
&= \int_{a-x}^{b-x} p'(u)\,du \\
&= p(b-x)-p(a-x), \qquad (47)
\end{aligned}$$

implying that $L_{xb}(x,a,b) \leq 0$ holds, which in turn proves $\hat{x}_b \geq 0$.

We next discuss the case where p is not necessarily differentiable. Since $p(u)$ is log-concave, $q(u) = -\log p(u)$ is convex, so that it is continuous and differentiable except on a countable set. Let $\phi(u)$ be any function such that $\phi(u)$ takes a value in the subderivative of $q(u)$ for any u. As $p(u) = e^{-q(u)}$, one has, at any point u at which $p(u)$ is differentiable,

$$p'(u) = -q'(u)e^{-q(u)} = -\phi(u)p(u). \qquad (48)$$

One therefore has, for any $b > a$,

$$p(b) - p(a) = -\int_a^b \phi(u)p(u)\,du, \qquad (49)$$

which can be proved in the same way as [Simon, 2011, Theorem 1.28]. On the other hand, we know that $\phi(u)$ is monotonically non-decreasing in u. One thus has

$$\phi(a)[P(b)-P(a)] = \phi(a)\int_a^b p(u)\,du \leq \int_a^b \phi(u)p(u)\,du \leq \phi(b)\int_a^b p(u)\,du = \phi(b)[P(b)-P(a)], \qquad (50)$$

which, via replacing (a, b) with $(a - x, b - x)$, proves inequalities corresponding to (46) and (47).

One has therefore proven the following proposition.

Proposition B.1. For $a < b$, let $L(x, a, b)$ be as defined in (39), and let $\hat{x}(a, b)$ be the minimizer of the optimization problem (38). Then, for $a < b$ and $a' < b'$ with $a \leq a'$ and $b \leq b'$, one has $\hat{x}(a, b) \leq \hat{x}(a', b')$.

We would like to note that if p is differentiable and *strictly* log-concave, then we have strict inequalities in the above proposition, which in turn implies, via (45), that $L_{xx}(x, a, b) \gtrsim 0$. This constitutes an alternative proof of the strict convexity of $L(x, a, b)$ in x under the differentiability assumption. Even without the differentiability assumption, one can note that, under the strict log-concavity of p, $\phi(u)$ is increasing, so that the strict inequalities hold in (50), proving the strict inequalities in the above proposition as well.

The above argument proves the first half of ONC3, that is, for any fixed w one has $z_q^* \leq z_{q+1}^*$ for $q \in \{1, \dots, Q-1\}$. Furthermore, if g' is strictly log-concave, one has the strict ordering when $w \neq 0$: $z_q^* < z_{q+1}^*$ for $q \in \{1, \dots, Q-1\}$, thereby proving the latter half of ONC3.

C Details of the Experimental Setup

Dataset statistics. We used the five publicly available real-world OR datasets of Gutiérrez et al. [2016][1]—ER, LE, and SW (Employee rejection/acceptance, Lecturers evaluation, and Social workers decisions (public domain) [Ben-David, 1992]), CA (Car evaluation (CC BY 4.0) [Bohanec and Rajkovič, 1988]), and WR (Wine quality—Red (CC BY 4.0) [Cortez et al., 2009])—exactly as released. The website offers 30 pre-defined training–validation hold-out splits whose label distributions are identical across the two partitions. We ran our experiments on all the 30 splits for each dataset and report the averages.

Table 1 summarizes the key statistics of the five datasets used in our study. For consistency, all ordinal labels were remapped to consecutive integers starting from one. We constructed input vectors by concatenating two types of preprocessed attributes: one-hot encoded categorical attributes and normalized numerical attributes.

Table 1: Summary of datasets used in the experiments. Attr. denotes attributes and Input dim. denotes the dimension of input vectors.

Dataset (code)	Subset	#Samples (per split)	#Attr.	Input dim.	#Classes (Q)	Distribution (counts per label)
ERA (ER)	Train	750	4	13	9	[1:69, 2:106, 3:136, 4:129, 5:118, 6:89, 7:66, 8:23, 9:14]
	Val	250				[1:23, 2:36, 3:45, 4:43, 5:40, 6:29, 7:22, 8:8, 9:4]
LEV (LE)	Train	750	4	9	5	[0:70, 1:210, 2:302, 3:148, 4:20]
	Val	250				[0:23, 1:70, 2:101, 3:49, 4:7]
SWD (SW)	Train	750	10	14	4	[2:24, 3:264, 4:299, 5:163]
	Val	250				[2:8, 3:88, 4:100, 5:54]
Car (CA)	Train	1296	6	21	4	[acc:288, good:52, unacc:907, vgood:49]
	Val	432				[acc:96, good:17, unacc:303, vgood:16]
Wine (WR)	Train	1199	11	17	6	[3:8, 4:39, 5:510, 6:479, 7:150, 8:13]
	Val	400				[3:2, 4:14, 5:171, 6:159, 7:49, 8:5]

Overparameterized network. Here we describe the neural network architecture used in our experiments.

- An input $\boldsymbol{x} \in \mathbb{R}^d$ is first mapped to a 128-dimensional representation by a linear layer, then passed through a parametric rectified linear unit (PReLU) [He et al., 2015].
- It then passes through four residual blocks, each defined as
$$\boldsymbol{x} \;\mapsto\; \boldsymbol{x} + \mathrm{PReLU}\big(W_2\, \mathrm{PReLU}(W_1 \boldsymbol{x} + \boldsymbol{b}_1) + \boldsymbol{b}_2\big), \quad W_1, W_2 \in \mathbb{R}^{128\times 128},\ \boldsymbol{b}_1, \boldsymbol{b}_2 \in \mathbb{R}^{128}.$$
- It subsequently passes through three consecutive linear layers with linear activation, yielding the 64-dimensional feature $\boldsymbol{h}_\theta(\boldsymbol{x})$.

[1] `https://www.uco.es/grupos/ayrna/orreview`.

- Finally, a linear layer without bias—whose weight vector is the classifier weight analyzed in this study—maps the $\boldsymbol{h}_\theta(\boldsymbol{x})$ to the one-dimensional latent variable $z = \boldsymbol{w}^\top \boldsymbol{h}_\theta(\boldsymbol{x})$.

The PReLU activations together with the linear tail give the network enough flexibility to map inputs to any location in the feature space, aligning with the UFM assumption. Figure 5 illustrates the architecture of the network.

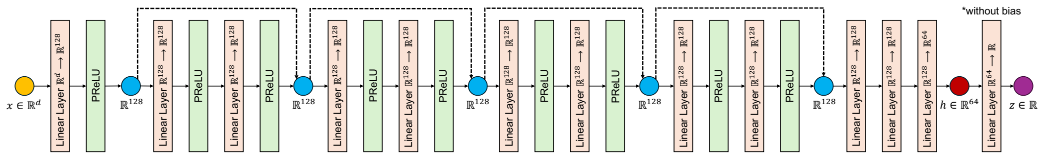

Figure 5: Architecture of the overparameterized network used in the experiments.

Other experimental settings. All the models were trained for 5000 epochs with the Adam optimizer, starting from the initial learning rates listed in Table 2. We made the learning rate to decay by a factor of 0.1 at epochs 200, 800, and 3000. We used a batch size of 2048 and applied a weight decay of 5×10^{-3} to all network parameters (except that the learnable thresholds, when present, received zero weight decay). For each dataset we tested the four possible combinations of the two link functions (logit, probit) and the two threshold strategies (fixed, learnable). Running every configuration on the 30 predefined hold-out splits yielded a total of 600 training–validation runs.

To ensure that the training error reached (near-)zero—the regime where feature collapse is observed—we tuned only the initial learning rate and, for fixed thresholds, the threshold range. Because the logistic function has heavier tails than the normal CDF, the logit runs used a wider fixed range $[-20, 20]$, whereas the probit runs used $[-2, 2]$. The complete hyper-parameter grid is summarized in Table 2.

Experiments compute resources. Every run was executed on a single NVIDIA RTX A6000 (48 GB) GPU. Each run took roughly 5–10 minutes and required only a few hundred megabytes of GPU memory. No multi-GPU or distributed training was used.

Table 2: Hyper-parameter settings of all experimental configurations. Each dataset was evaluated under two link functions (logit, probit) and two threshold strategies (fixed, learnable).

Dataset	Link function	Thresholds	Threshold range (when fixed)	Initial learning rate
ER	logit	fixed	$[-20,\ 20]$	1×10^{-2}
	logit	learnable	–	1×10^{-2}
	probit	fixed	$[-2,\ 2]$	1×10^{-3}
	probit	learnable	–	1×10^{-3}
LE	logit	fixed	$[-20,\ 20]$	1×10^{-2}
	logit	learnable	–	1×10^{-2}
	probit	fixed	$[-2,\ 2]$	5×10^{-3}
	probit	learnable	–	5×10^{-3}
SW	logit	fixed	$[-20,\ 20]$	1×10^{-2}
	logit	learnable	–	1×10^{-2}
	probit	fixed	$[-2,\ 2]$	5×10^{-3}
	probit	learnable	–	5×10^{-3}
CA	logit	fixed	$[-20,\ 20]$	1×10^{-2}
	logit	learnable	–	1×10^{-2}
	probit	fixed	$[-2,\ 2]$	5×10^{-3}
	probit	learnable	–	5×10^{-3}
WR	logit	fixed	$[-20,\ 20]$	1×10^{-2}
	logit	learnable	–	1×10^{-2}
	probit	fixed	$[-2,\ 2]$	1×10^{-3}
	probit	learnable	–	1×10^{-3}

D Additional experimental results

D.1 Results with the logit model

This section presents the experimental outcomes obtained using the logistic function (i.e., $g(x) = (1 + e^{-x})^{-1}$), which corresponds to the logit model. Figures 6–9 show the evolution of evaluation-metric curves for the datasets LE, SW, CA, and WR, respectively. Figures 10–13 show visualization of the latent and feature spaces for the datasets LE, SW, CA, and WR, respectively. The results for the dataset ER are presented as Figs. 2 and 3 in the main text.

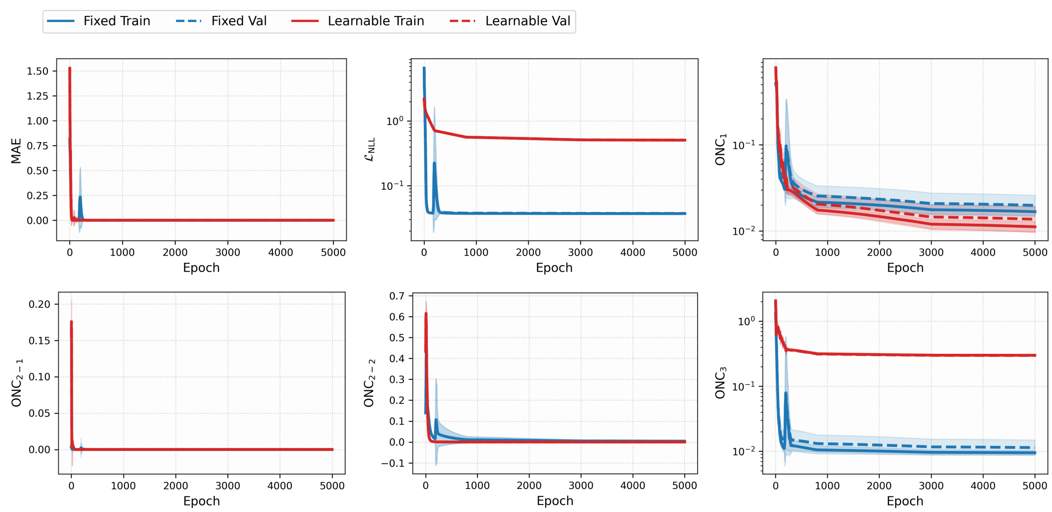

Figure 6: Epoch-wise average metrics curves for the LE dataset with the logit model, comparing fixed- and learnable-threshold models.

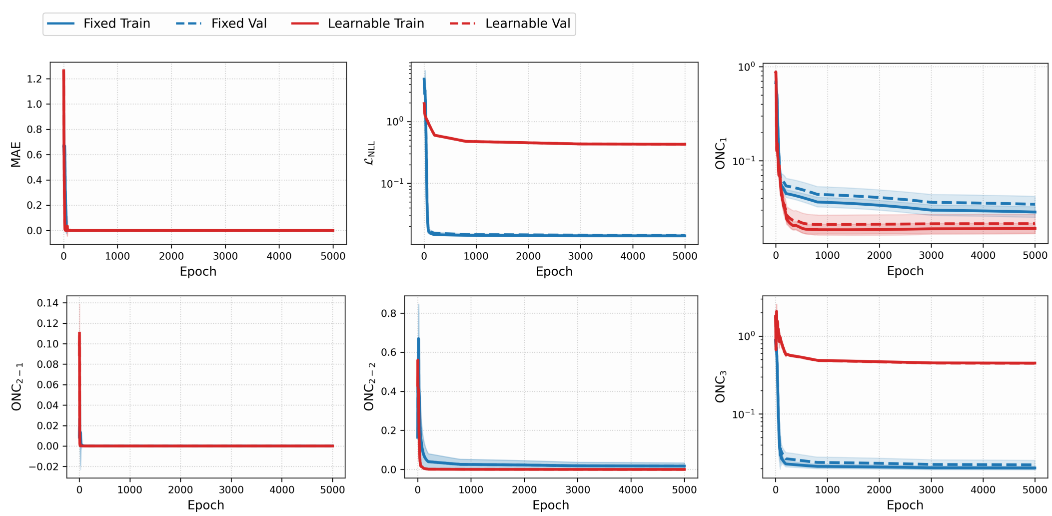

Figure 7: Epoch-wise average metrics curves for the SW dataset with the logit model, comparing fixed- and learnable-threshold models.

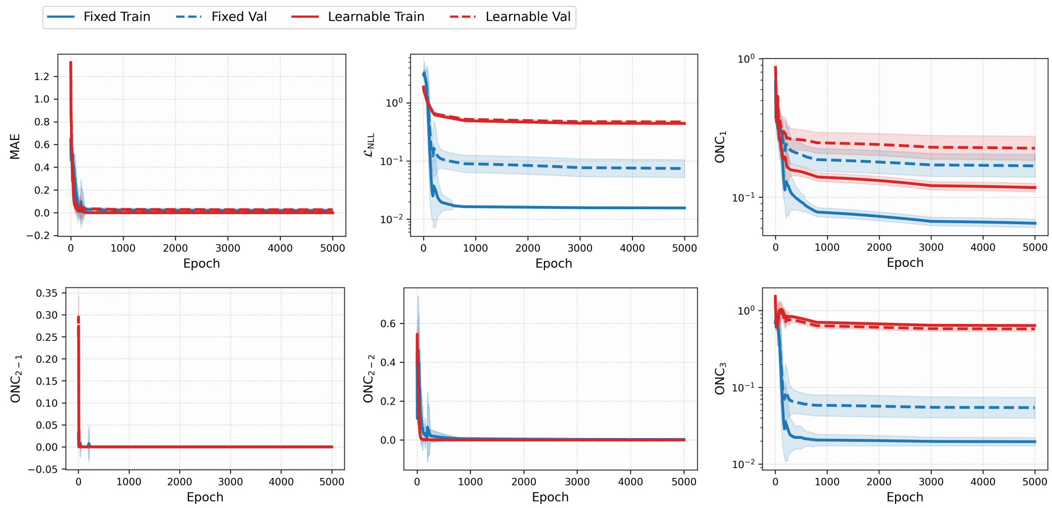

Figure 8: Epoch-wise average metrics curves for the CA dataset with the logit model, comparing fixed- and learnable-threshold models.

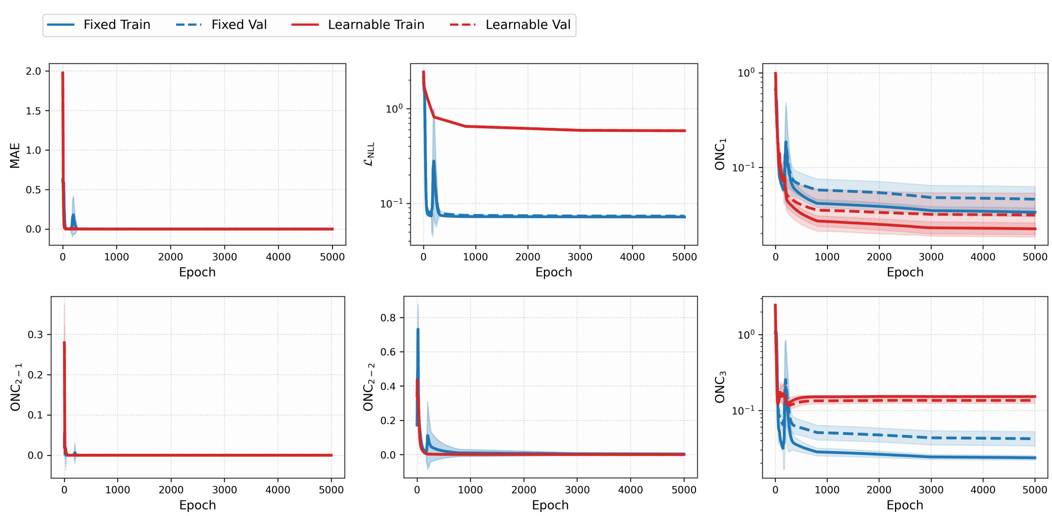

Figure 9: Epoch-wise average metrics curves for the WR dataset with the logit model, comparing fixed- and learnable-threshold models.

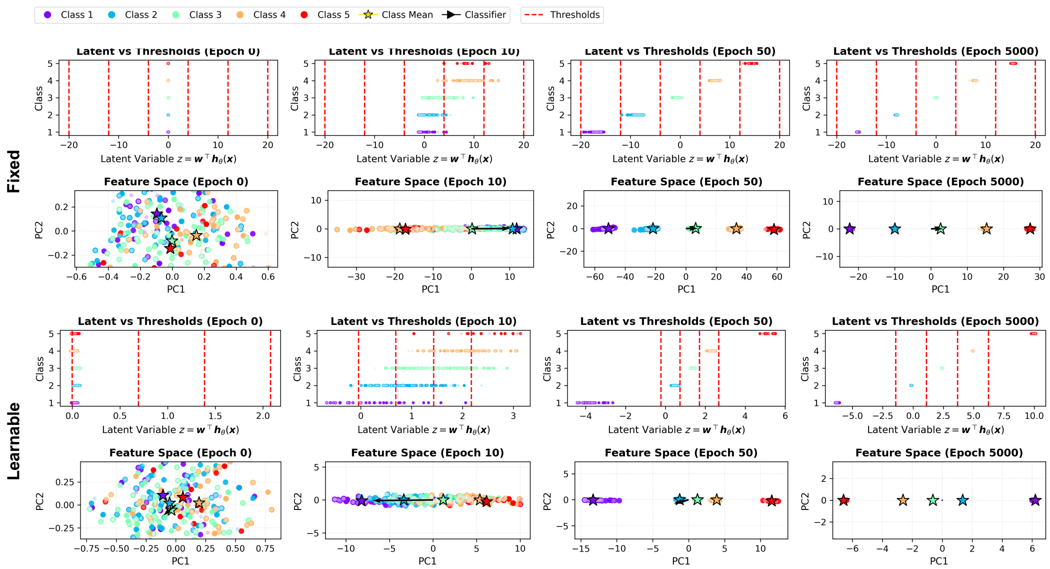

Figure 10: Visualization of the latent and feature spaces for the LE dataset using the logit model, comparing fixed- and learnable-threshold models.

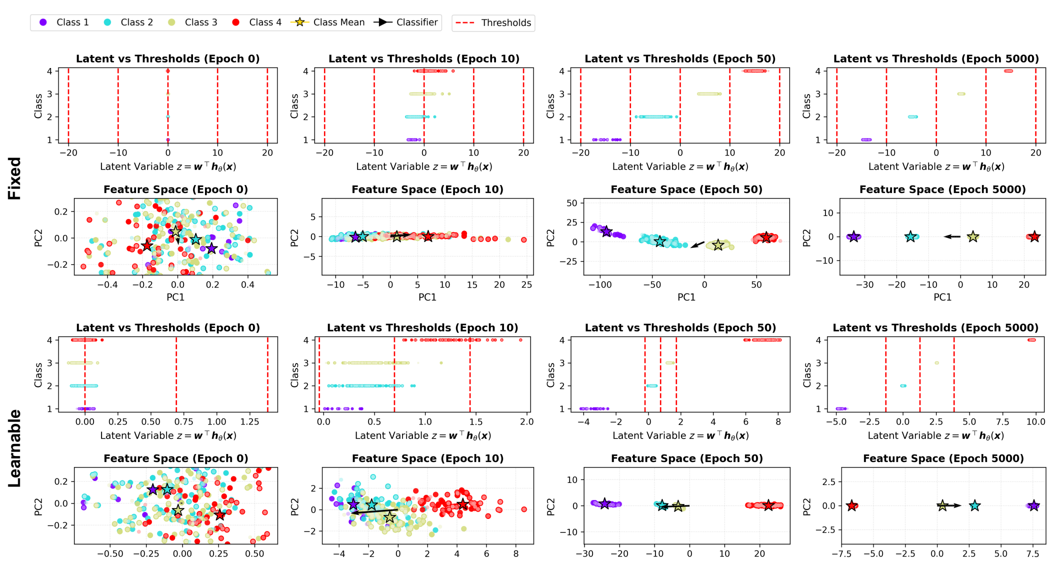

Figure 11: Visualization of the latent and feature spaces for the SW dataset using the logit model, comparing fixed- and learnable-threshold models.

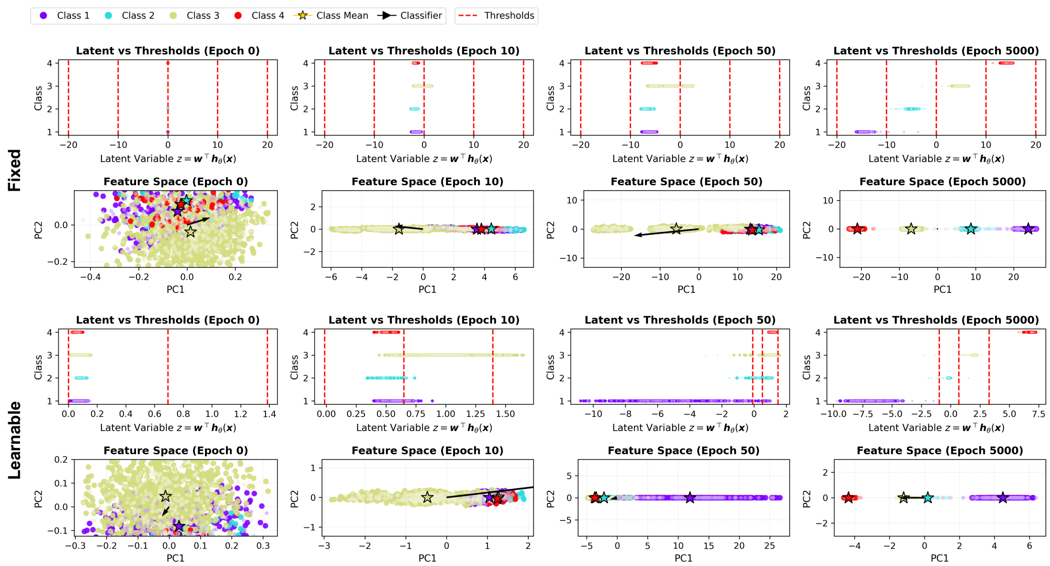

Figure 12: Visualization of the latent and feature spaces for the CA dataset using the logit model, comparing fixed- and learnable-threshold models.

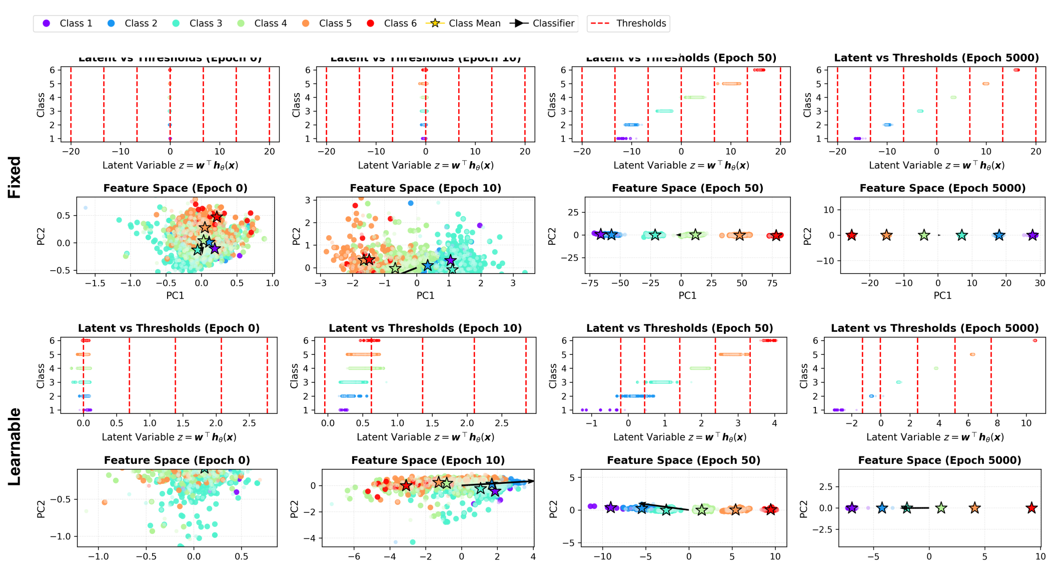

Figure 13: Visualization of the latent and feature spaces for the WR dataset using the logit model, comparing fixed- and learnable-threshold models.

D.2 Results with the probit model

This section presents the experimental outcomes obtained using the normal CDF (i.e., $g(x) = \Phi(x)$), which corresponds to the probit model. Figures 14–18 show the evolution of evaluation-metric curves for the datasets ER, LE, SW, CA, and WR, respectively. Figures 19–23 show visualization of the latent and feature spaces for the datasets ER, LE, SW, CA, and WR, respectively.

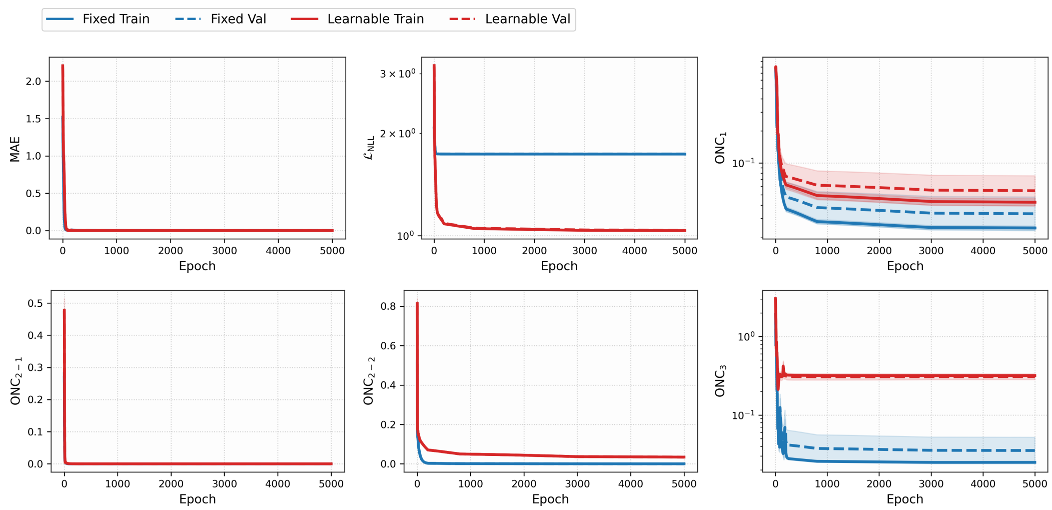

Figure 14: Epoch-wise average metrics curves for the ER dataset with the probit model, comparing fixed- and learnable-threshold models.

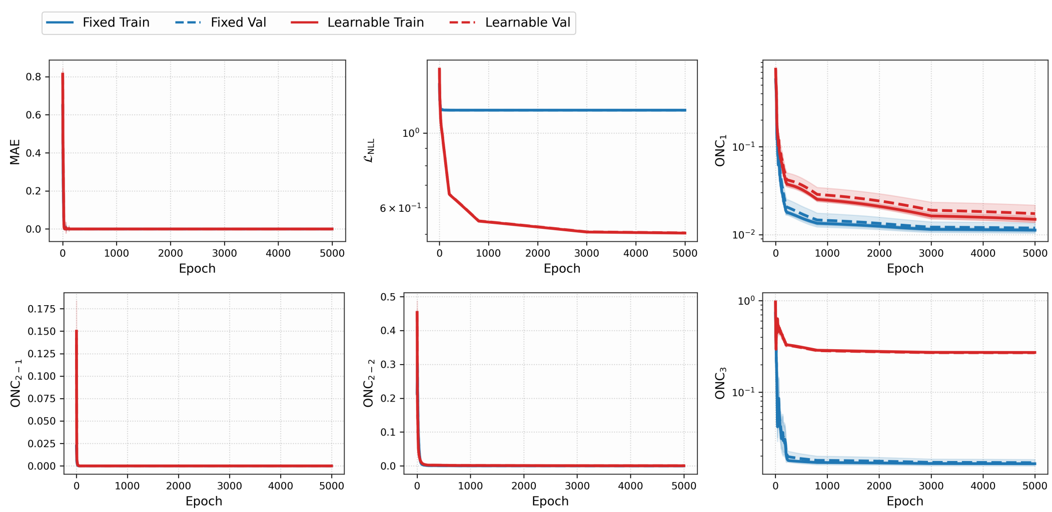

Figure 15: Epoch-wise average metrics curves for the LE dataset with the probit model, comparing fixed- and learnable-threshold models.

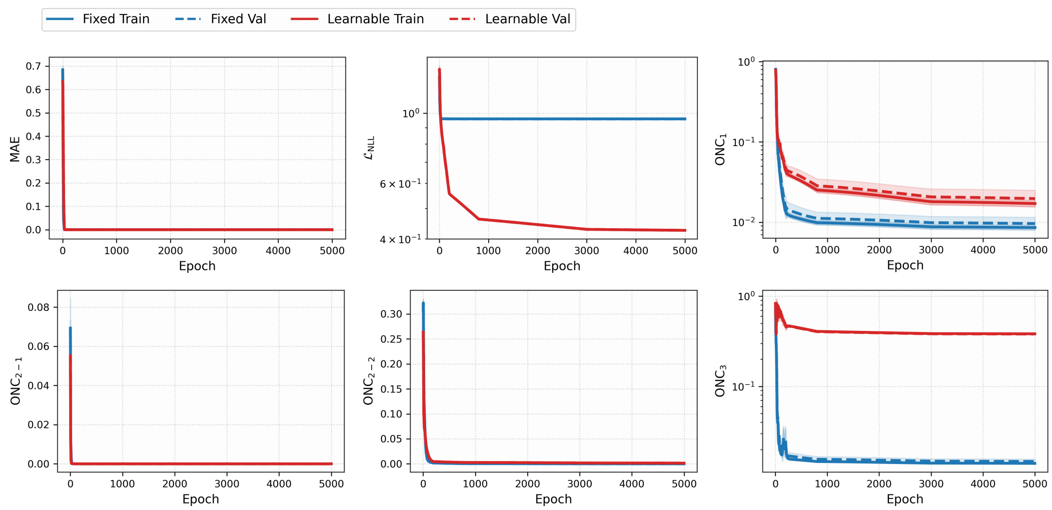

Figure 16: Epoch-wise average metrics curves for the SW dataset with the probit model, comparing fixed- and learnable-threshold models.

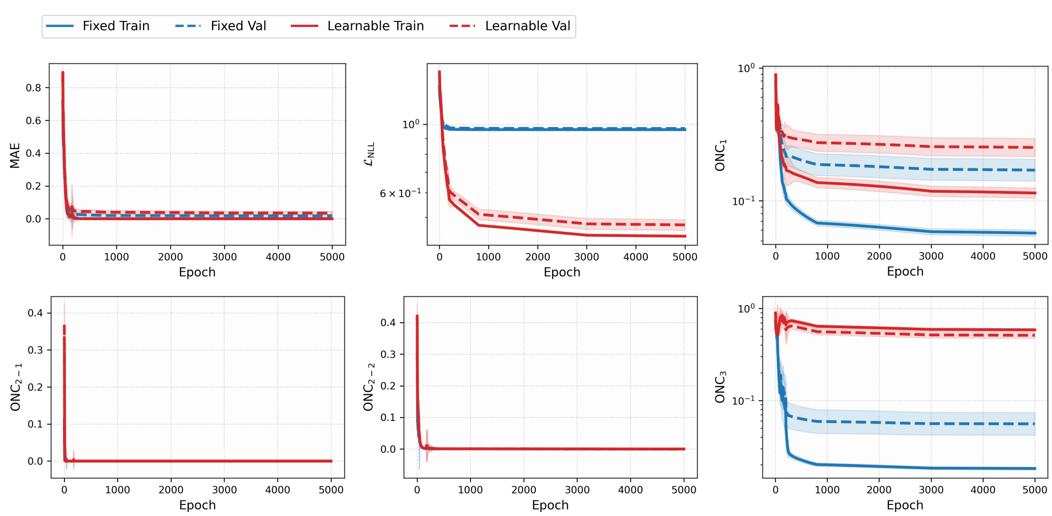

Figure 17: Epoch-wise average metrics curves for the CA dataset with the probit model, comparing fixed- and learnable-threshold models.

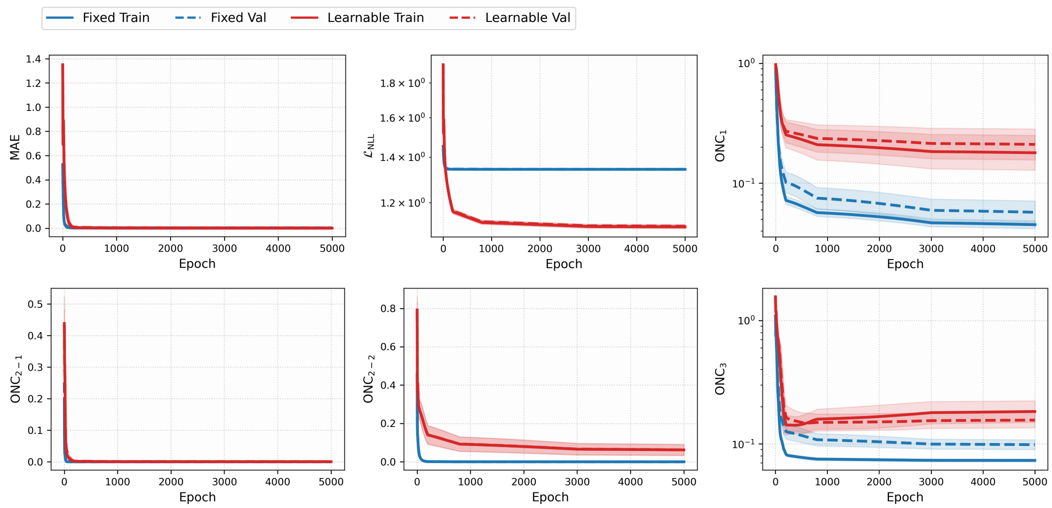

Figure 18: Epoch-wise average metrics curves for the WR dataset with the probit model, comparing fixed- and learnable-threshold models.

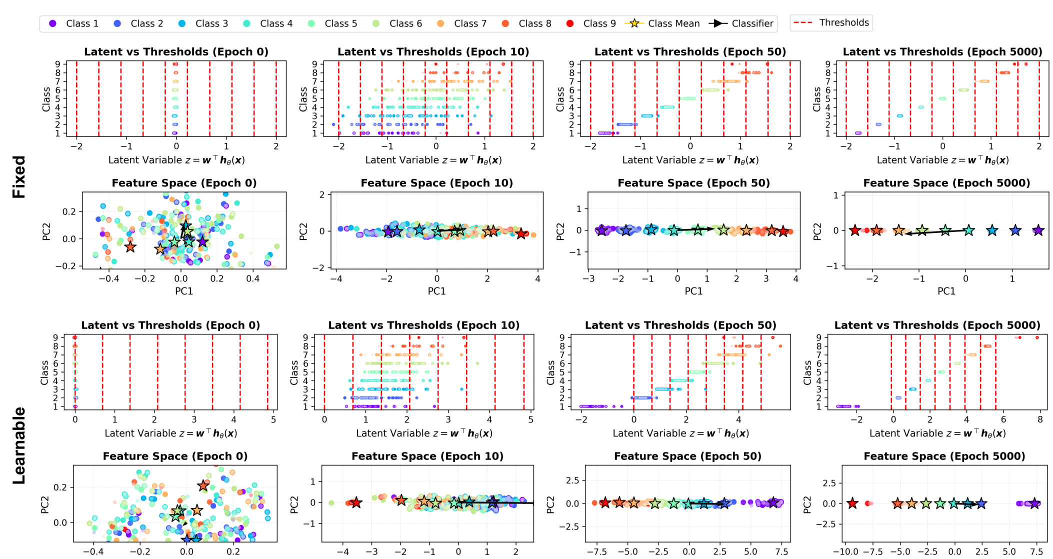

Figure 19: Visualization of the latent and feature spaces for the ER dataset using the probit model, comparing fixed- and learnable-threshold models.

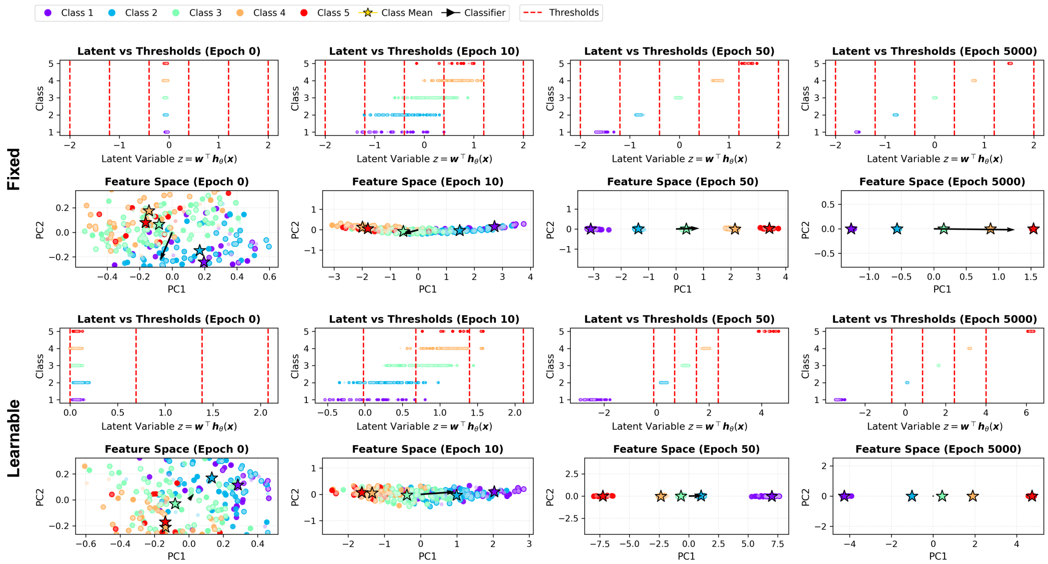

Figure 20: Visualization of the latent and feature spaces for the LE dataset using the probit model, comparing fixed- and learnable-threshold models.

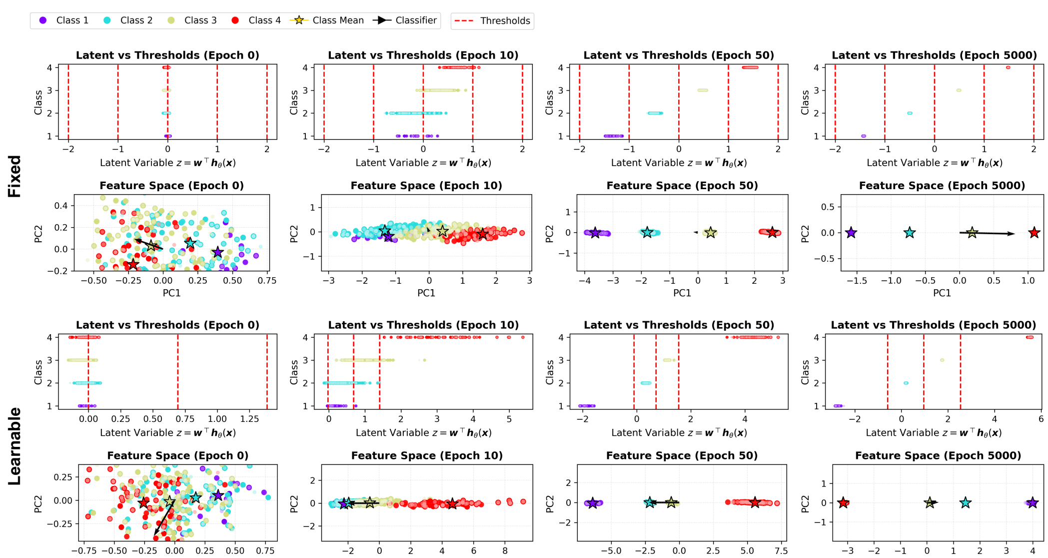

Figure 21: Visualization of the latent and feature spaces for the SW dataset using the probit model, comparing fixed- and learnable-threshold models.

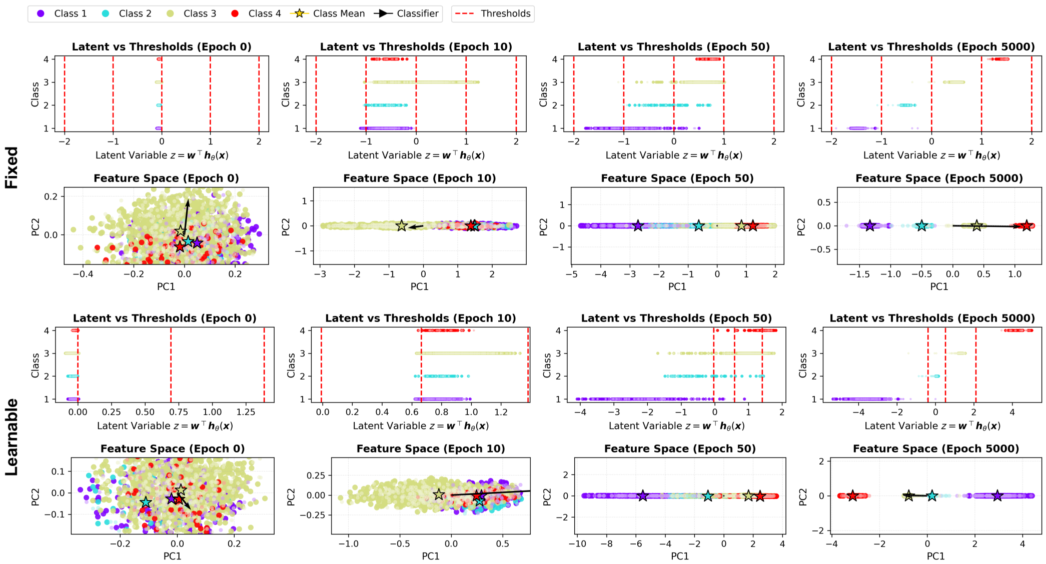

Figure 22: Visualization of the latent and feature spaces for the CA dataset using the probit model, comparing fixed- and learnable-threshold models.

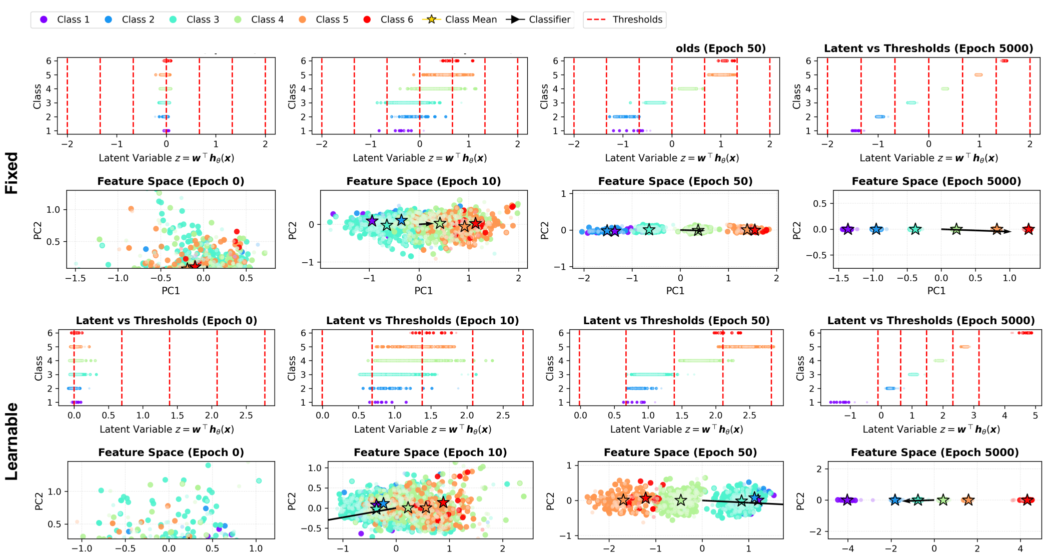

Figure 23: Visualization of the latent and feature spaces for the WR dataset using the probit model, comparing fixed- and learnable-threshold models.